

Project case study

Telefónica: building a global cloud foundation on AWS

A unified, cloud-native platform now serves 46,000 end-user devices across 750 locations, giving Telefónica a secure base for SaaS delivery, data migration and continued growth.

46,000

end-user devices supported

750

locations connected

Multi-region

countries and time zones, one operating model

Why change was needed

Telefónica's legacy infrastructure could no longer keep pace with rapid growth, distributed operations and shifting customer demands.

The Global Infrastructure Transformation Programme responded with a cloud-first architecture built on Amazon Web Services (AWS). The aim was elasticity, resilience and cost efficiency across the whole enterprise, with core infrastructure modernised, SaaS adoption accelerated and data migrated securely at scale.

What the programme delivered

Work ran across five connected areas, from the AWS foundation through to day-to-day service management.

AWS infrastructure modernisation

- Designed and deployed a **multi-region AWS environment** for compute, storage and networking workloads.
- Adopted **Infrastructure-as-Code** with Terraform and CloudFormation for repeatable, compliant provisioning.
- Set up automated monitoring, patching and backup frameworks aligned to **ISO 27001 and GDPR**.

SaaS development and integration

- Delivered new internal and customer-facing SaaS applications on **microservices and containerised** architectures.
- Integrated identity, billing and analytics modules using AWS Lambda, API Gateway and DynamoDB.
- Enabled **CI/CD pipelines** with AWS CodePipeline and GitHub Actions.

Data migration and cloud storage

- Moved structured and unstructured data from on-premises systems to **AWS S3, RDS and Redshift**.
- Protected integrity with checksum validation, encryption at rest and in transit, and automated reconciliation.
- Introduced a global data-replication strategy to keep performance consistent across time zones.

End-user estate transformation

- Supported **46,000 devices across 750 locations** with secure connectivity, identity federation and endpoint management.
- Applied unified access policies and device compliance through AWS WorkSpaces and Intune integration.
- Provided early-life support and user adoption campaigns for a smooth transition.

Operational governance and service management

- Integrated **ITSM with ServiceNow** for incident, change and configuration management.
- Authored service designs, operational manuals and runbooks for cloud operations.
- Ran training and knowledge-transfer sessions for internal teams in **Europe, Latin America and Asia**.

Outcomes

- A global AWS infrastructure supporting 46,000 users across 750 locations.
- A modern SaaS ecosystem that speeds up product delivery and innovation.
- Secure, compliant data migration with better performance and resilience.
- One operational model that works across multiple time zones.
- Lower infrastructure costs and improved scalability.

Strategic impact

The programme gave Telefónica a global digital backbone that connects people, platforms and data across continents. The result is a future-ready environment built for continuous innovation, operational agility and sustainable growth.

Connecting people, platforms and data across continents.

Technology at a glance

AWS S3, RDS, Redshift, Lambda, API Gateway, DynamoDB, CodePipeline, WorkSpaces; GitHub Actions; Terraform and CloudFormation; Intune; ServiceNow.